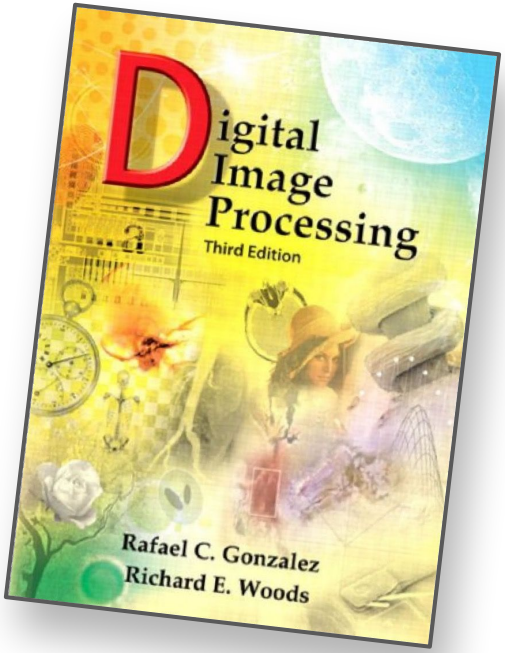
Sistema Visual Humano

Agenda

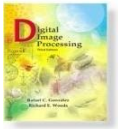
- Estructura del Ojo Humano
- Formación de Imágenes
- Intensidad Percibida
- La luz y el espectro electromagnético

Nota

- Esta sesión está basada en el libro “Digital Image Processing” 3ra edición de Rafael C. González y Richard E. Woods. En especial el capítulo 2.



Estructura del ojo humano



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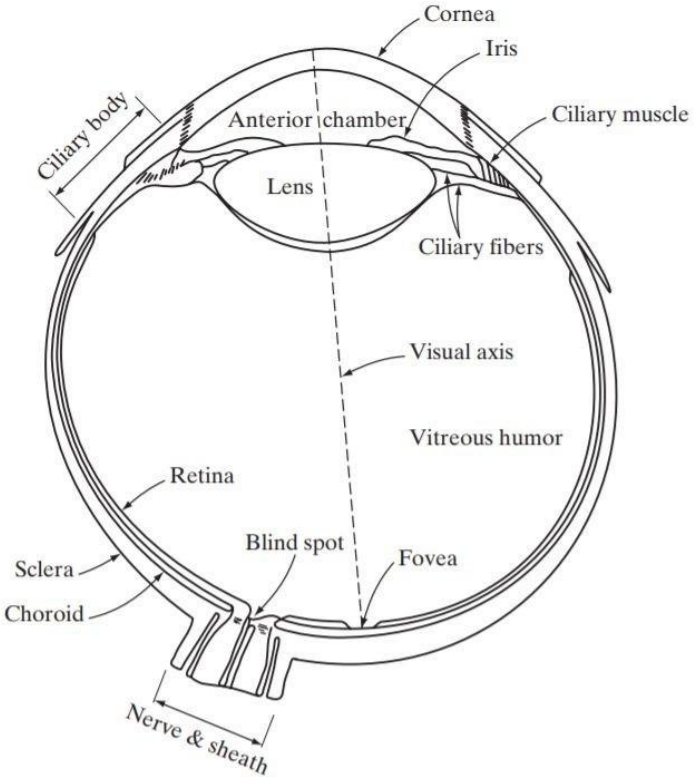
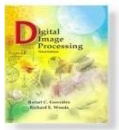


FIGURE 2.1 Simplified diagram of a cross section of the human eye.

Estructura del ojo humano



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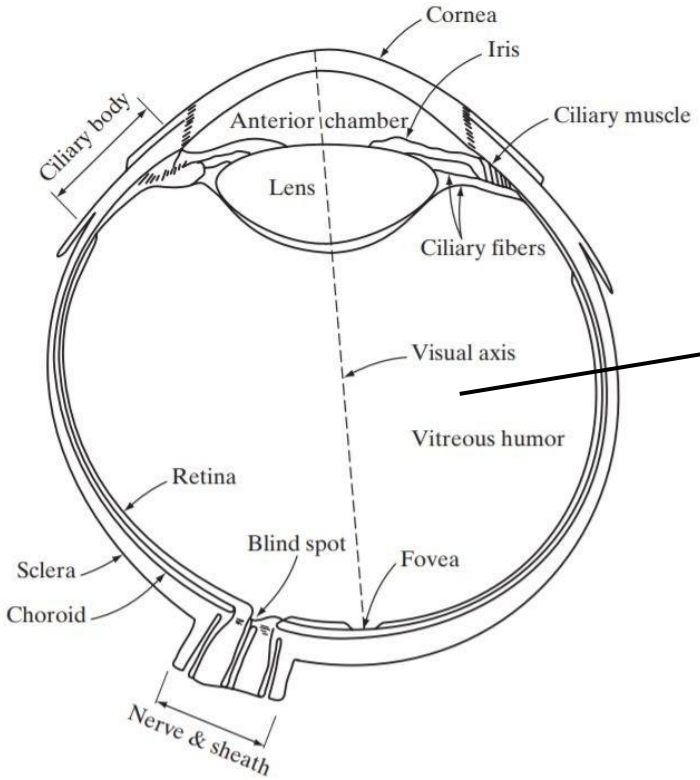
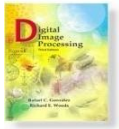


FIGURE 2.1 Simplified diagram of a cross section of the human eye.

El ojo humano es aproximadamente una esfera con un diámetro de 20mm

Estructura del ojo humano



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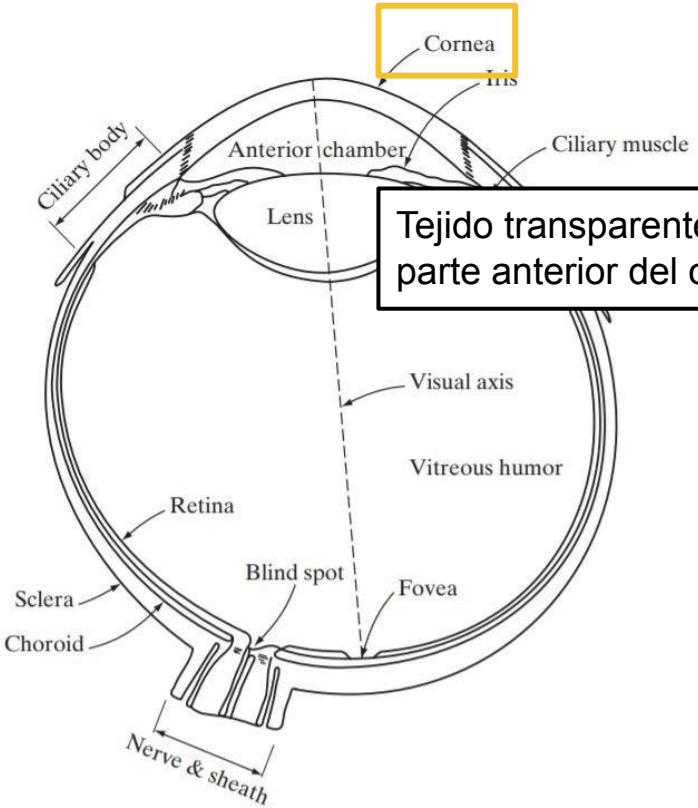


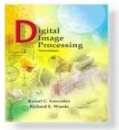
FIGURE 2.1 Simplified diagram of a cross section of the human eye.

Tejido transparente y fuerte que cubre la parte anterior del ojo

Tres membranas delimitan el ojo:

- Cornea

Estructura del ojo humano



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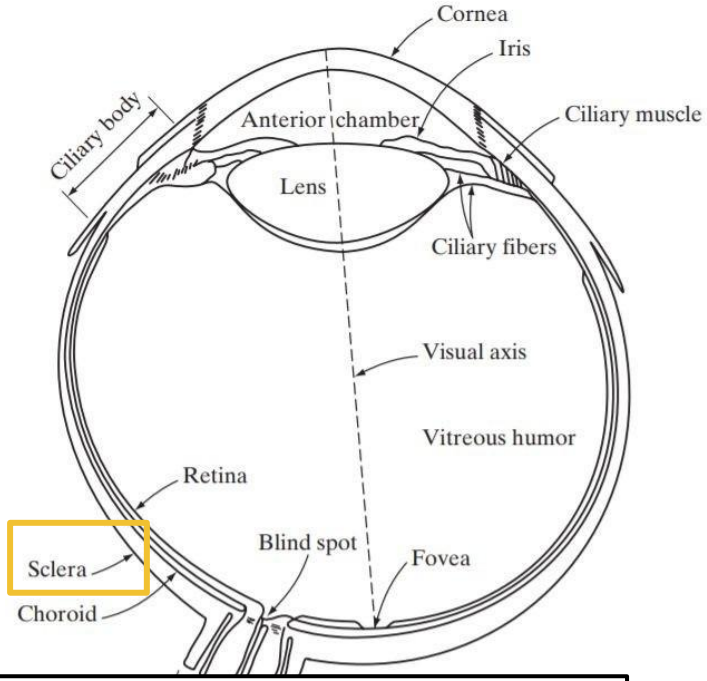

FIGURE 2.1 Simplified diagram of a cross section of the human eye.

Membrana opaca que cubre el resto del nervio óptico

Tres membranas delimitan el ojo:

- Cornea y esclera (cobertura exterior)

Estructura del ojo humano

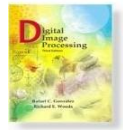

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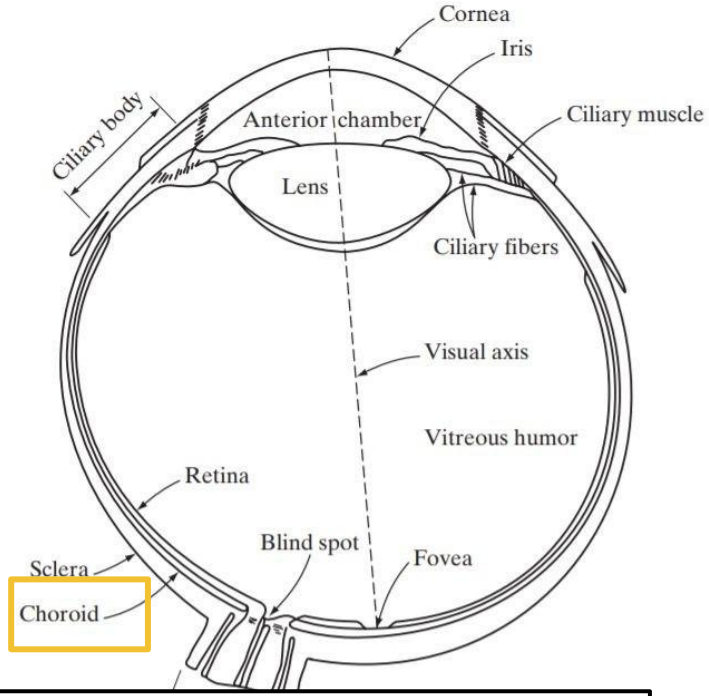
FIGURE 2.1 Simplified diagram of a cross section of the human eye.

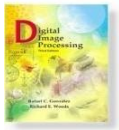


Contiene una red de vasos capilares que son la fuente más importante de nutrición del ojo

Tres membranas delimitan el ojo:

- Cornea y esclera (cobertura exterior)
- Choroid

Estructura del ojo humano



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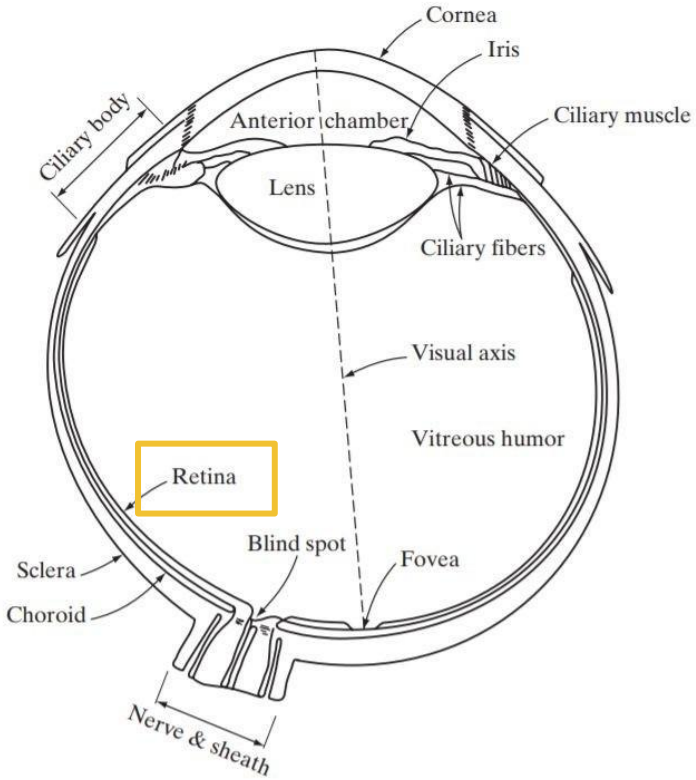
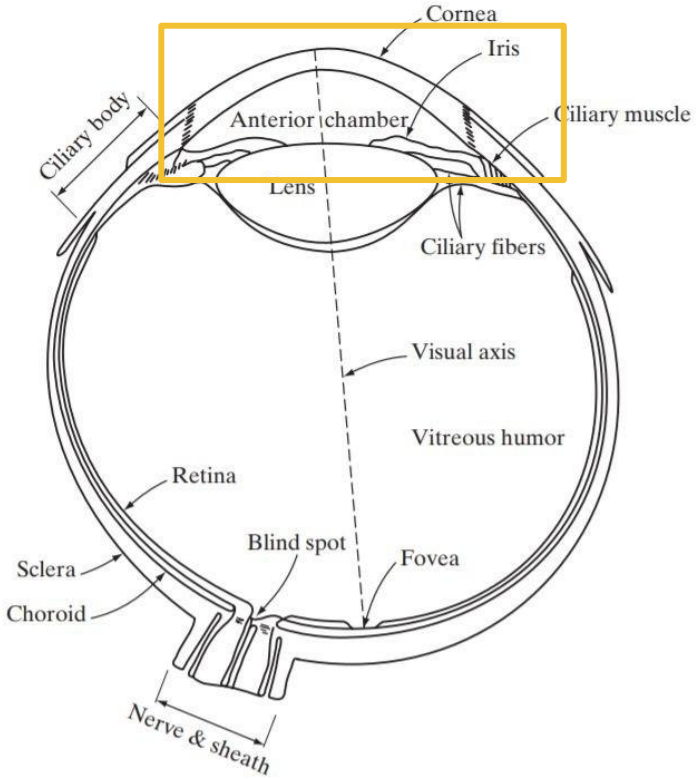


FIGURE 2.1 Simplified diagram of a cross section of the human eye.

Tres membranas delimitan el ojo:

- Cornea y esclera (cobertura exterior)
- Choroid
- Retina

Estructura del ojo humano



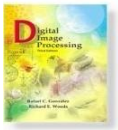


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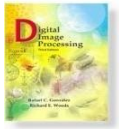
FIGURE 2.1 Simplified diagram of a cross section of the human eye.

- El iris se contrae o expande para controlar la cantidad de luz que entra al ojo
- La apertura central del iris (la pupila) varía en diámetro aprox. desde 2-8mm

Estructura del ojo humano



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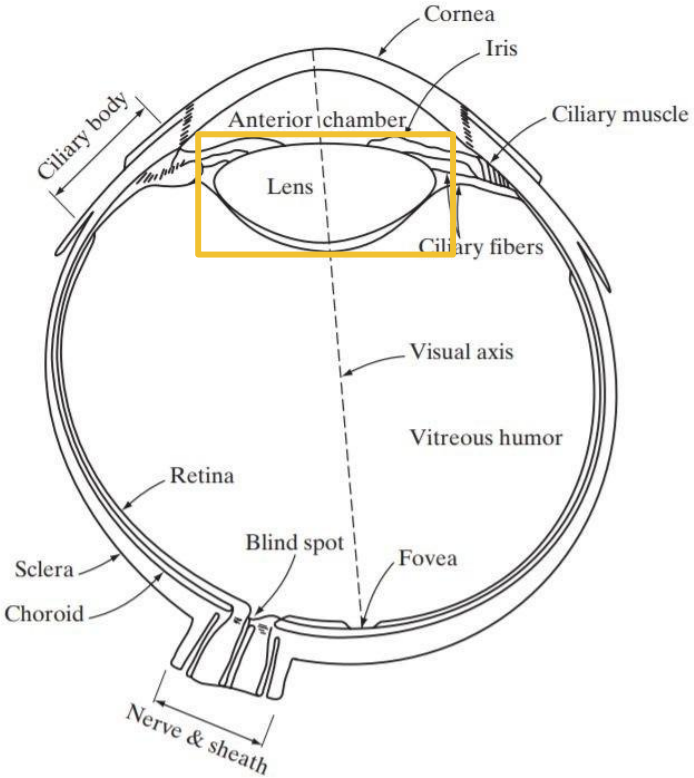
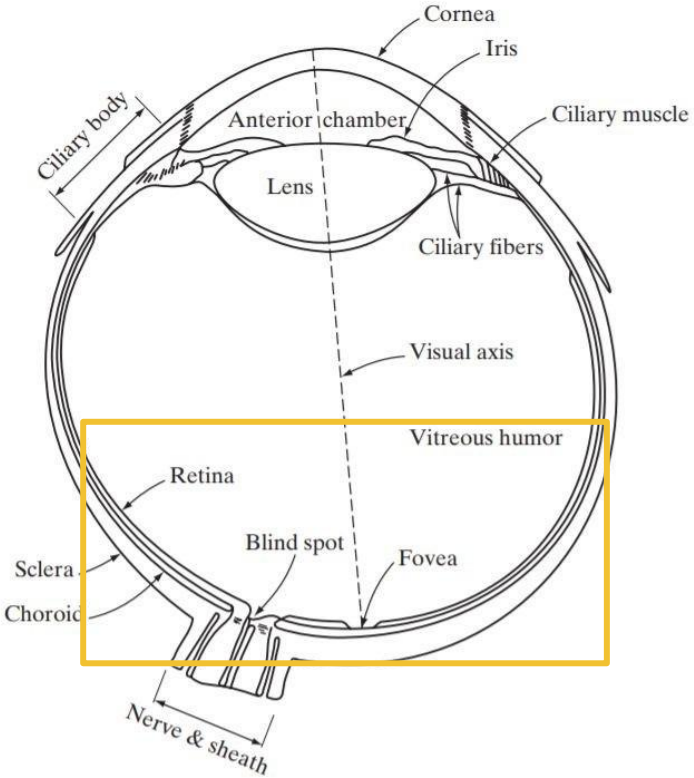


FIGURE 2.1 Simplified diagram of a cross section of the human eye.

- El lente está compuesto por capas concéntricas de células fibrosas
- Contiene 60% a 70% de agua, 6% grasa y proteína
- Puede ser afectada por cataratas
- Absorbe aprox. el 8% de la luz del espectro visible

Estructura del ojo humano



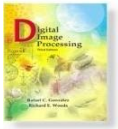


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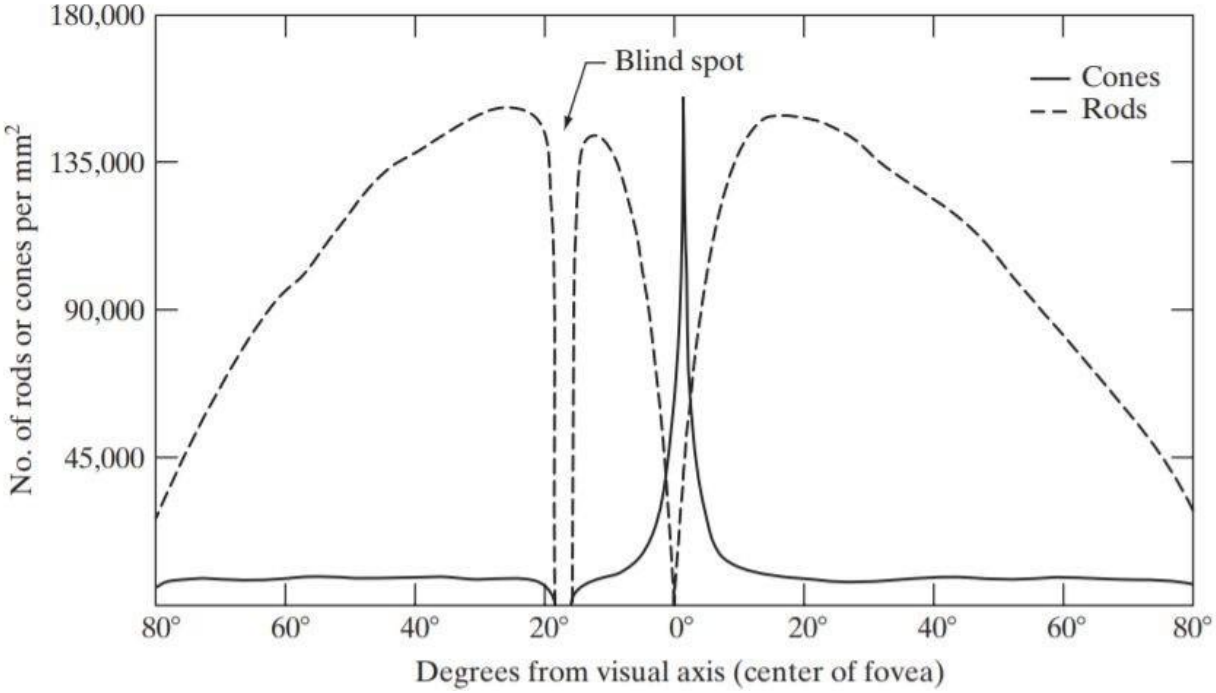
FIGURE 2.1 Simplified diagram of a cross section of the human eye.

- Cuando una imagen está apropiadamente enfocada, su imagen se encuentra en la retina
- Receptores de luz
  - Conos 6-7 millones (visión de color o fotópica /bright-light vision)
  - Bastonetes (75-150 millones) -- dan una idea general de la escena

Distribución de los conos y bastones

FIGURE 2.2 Distribution of rods and cones in the retina.



Formación de Imágenes en el Ojo

- En una cámara fotográfica, el lente tiene una longitud focal fija. Se consigue el enfoque al variar la distancia entre el lente y el plano del sensor
- En el ojo humano es al revés, la distancia entre el lente y la retina es fija
  - La longitud focal necesaria para enfocar es obtenida al cambiar la forma del lente
- El rango de las longitudes focales es de 14mm a 17mm (cuando el ojo está

  relajado y se enfoca a 3m)

Formación de Imágenes en el Ojo

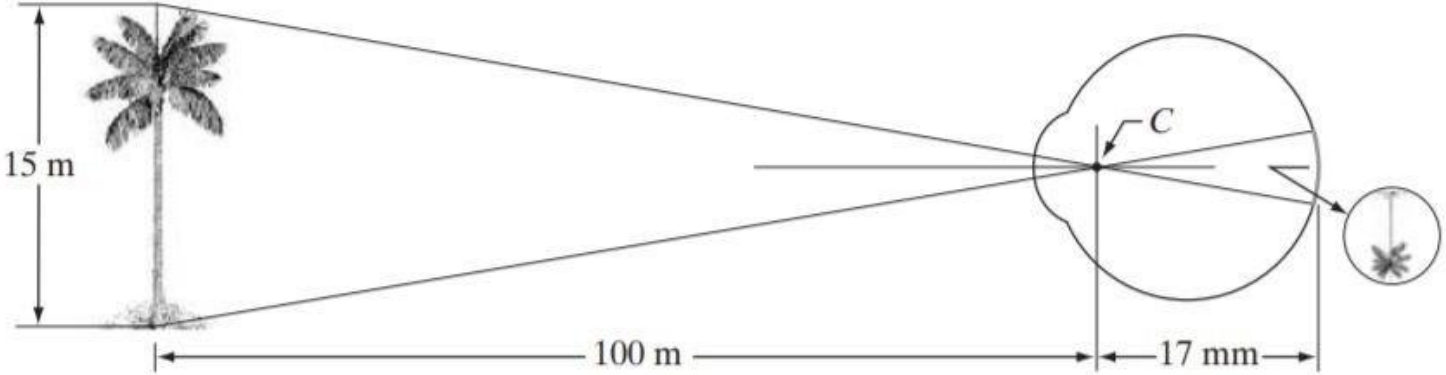
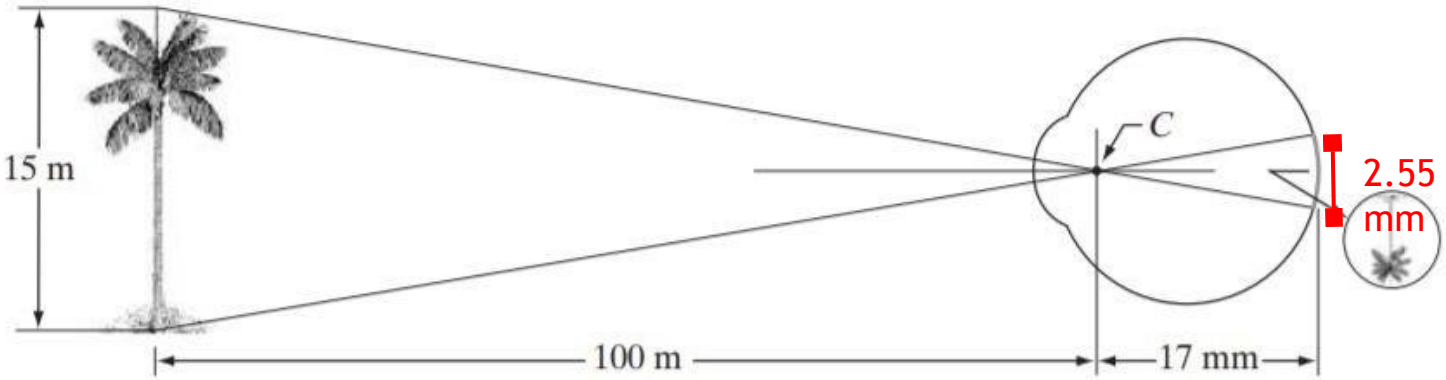


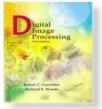
FIGURE 2.3 Graphical representation of the eye looking at a palm tree. Point C is the optical center of the lens.

Formación de Imágenes en el Ojo



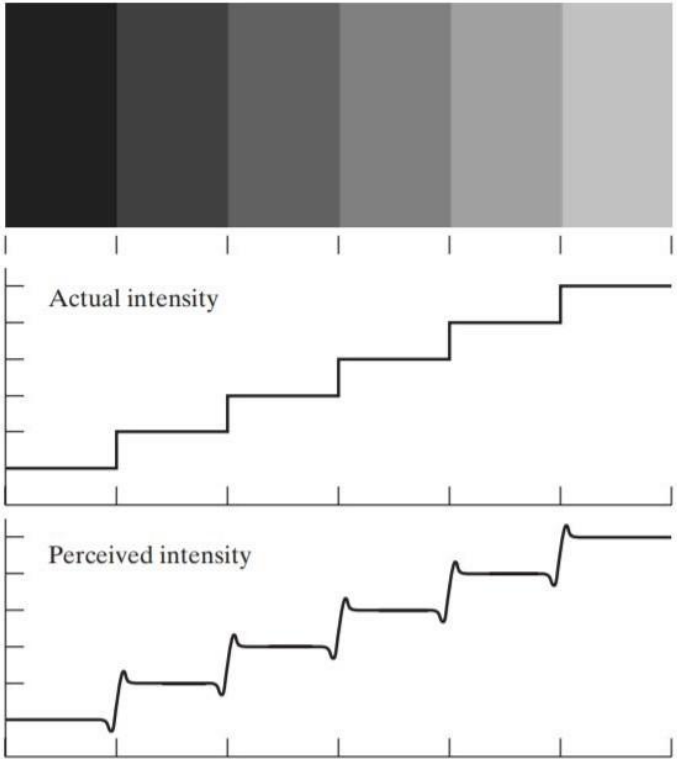
15/100 = h/17
h= 2.55 mm
Nota: la imagen está invertida!

Intensidad percibida

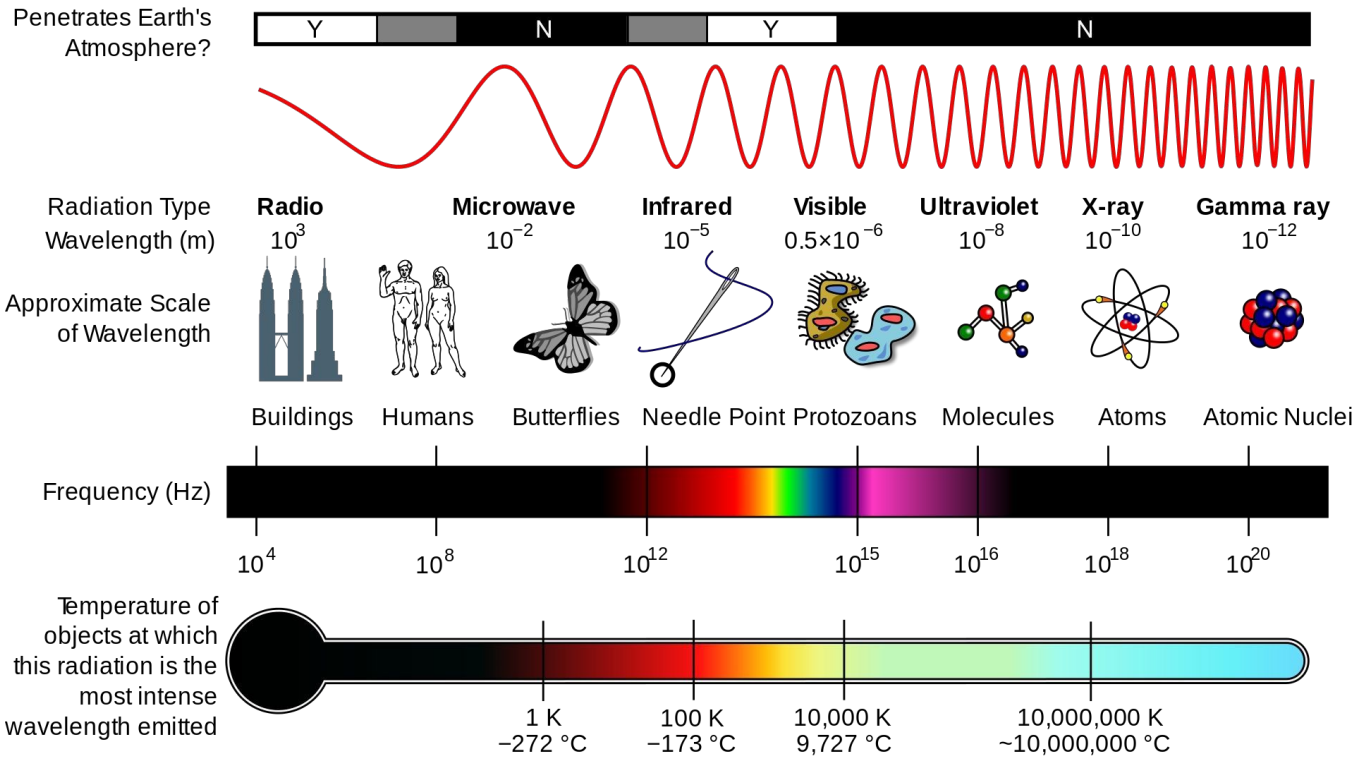


a
b
c

FIGURE 2.7 Illustration of the Mach band effect. Perceived intensity is not a simple function of actual intensity.



La luz y el espectro electromagnético



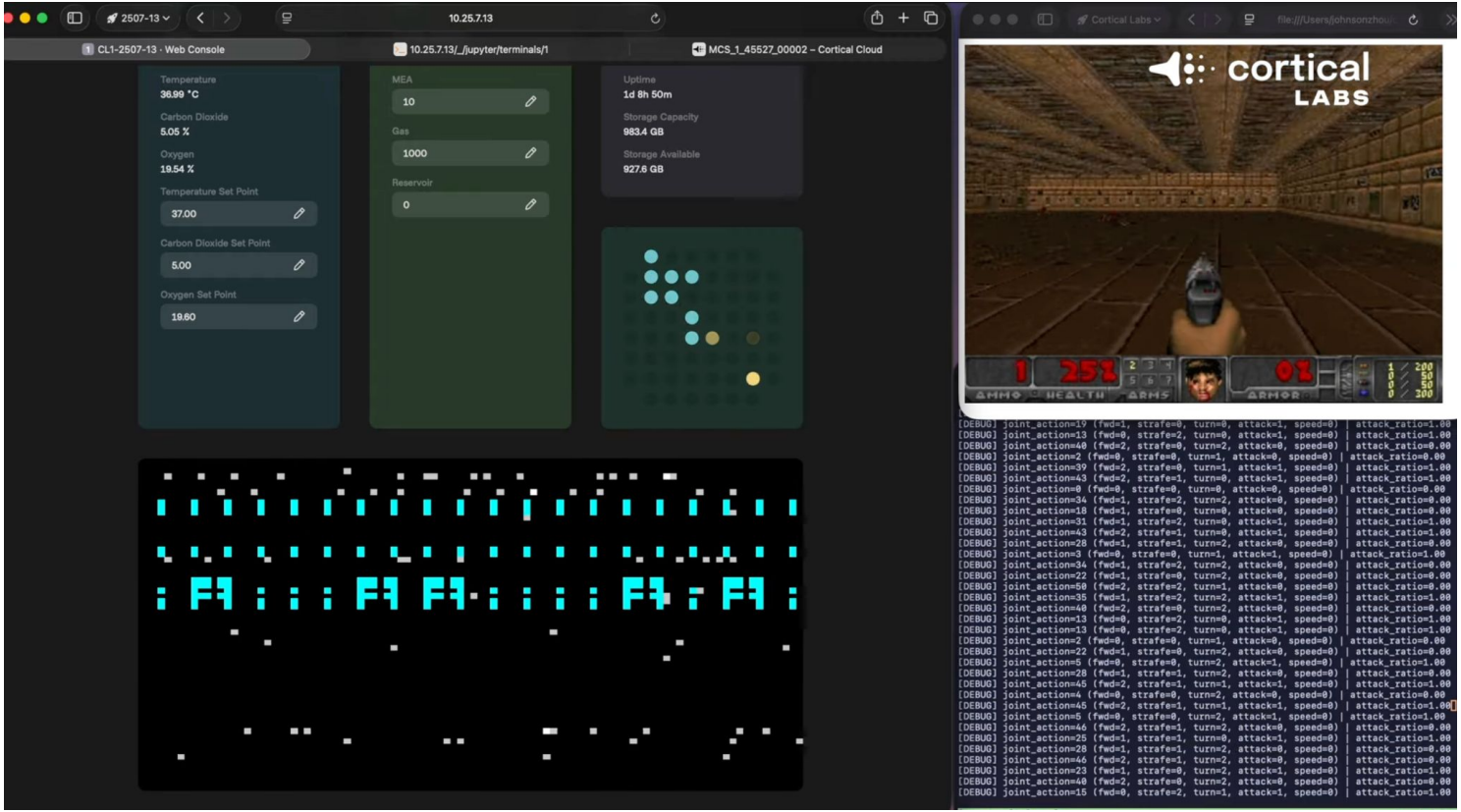
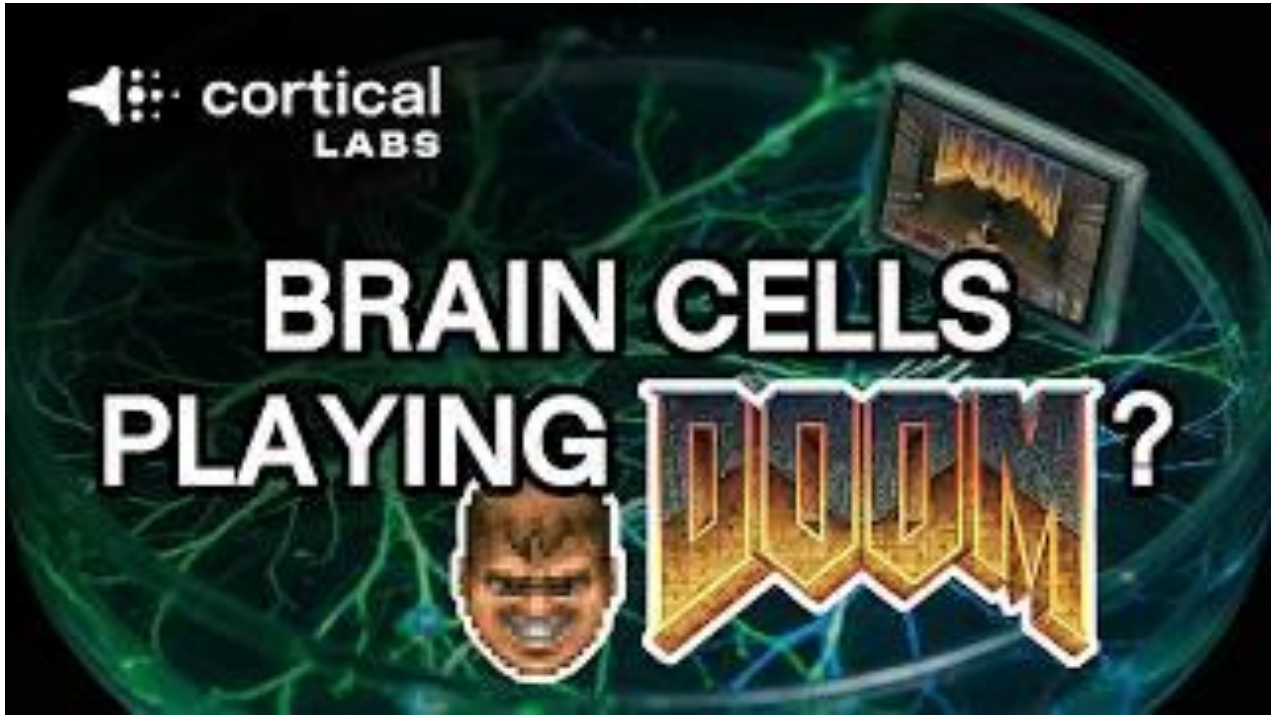
https://en.wikipedia.org/wiki/Electromagnetic_spectrum#/media/File:EM_Spectrum_Properties_edit.svg

Otros tipos de sistemas visuales

Perros y gatos: 2 tipos de receptores de color

Camarón Mantis: 16 tipos de receptores de color

Neuronas cultivadas en Cortical Labs CL-1



Neuronas cultivadas en Cortical Labs CL-1